

14-Dec-2024

Back propagation



$$x_1 \rightarrow w, b \rightarrow 0,$$

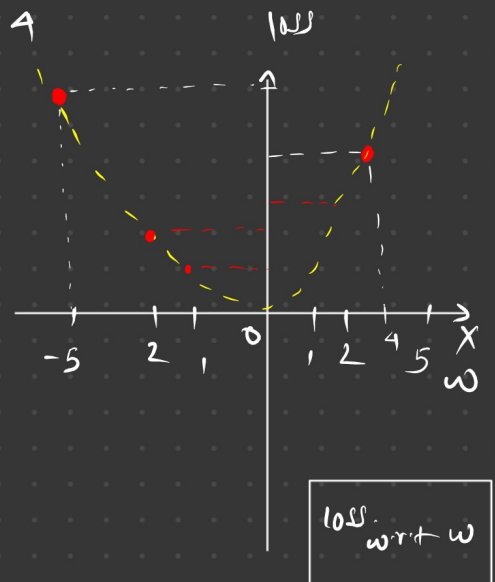
$$0.5 \rightarrow w, b \rightarrow 1.6 \quad \text{outcome}$$

$$\rightarrow 2 \quad \text{Actual}$$

$$\text{loss} \rightarrow (\text{outcome} - \text{Actual})$$

$$w \rightarrow w_1 \rightarrow -5$$

$$w \rightarrow w_2 \rightarrow 4$$

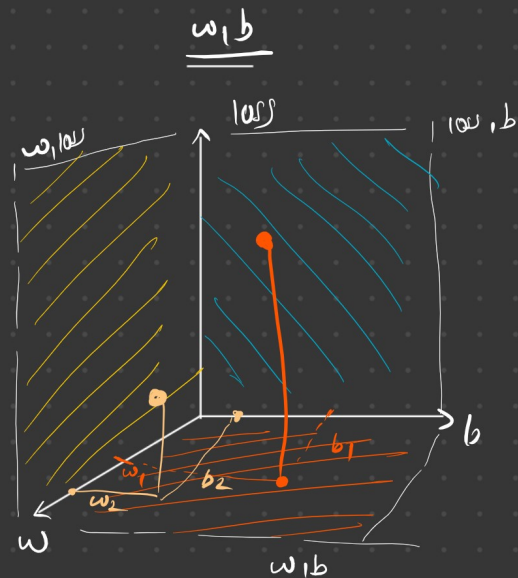
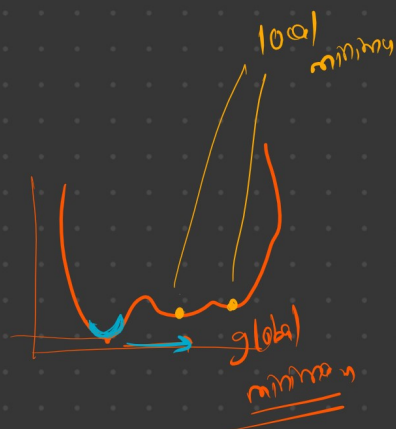


$\rightarrow \text{minimize loss} \downarrow$

$$x_1 \cdot w + b = 2 \rightarrow \text{Actual}$$

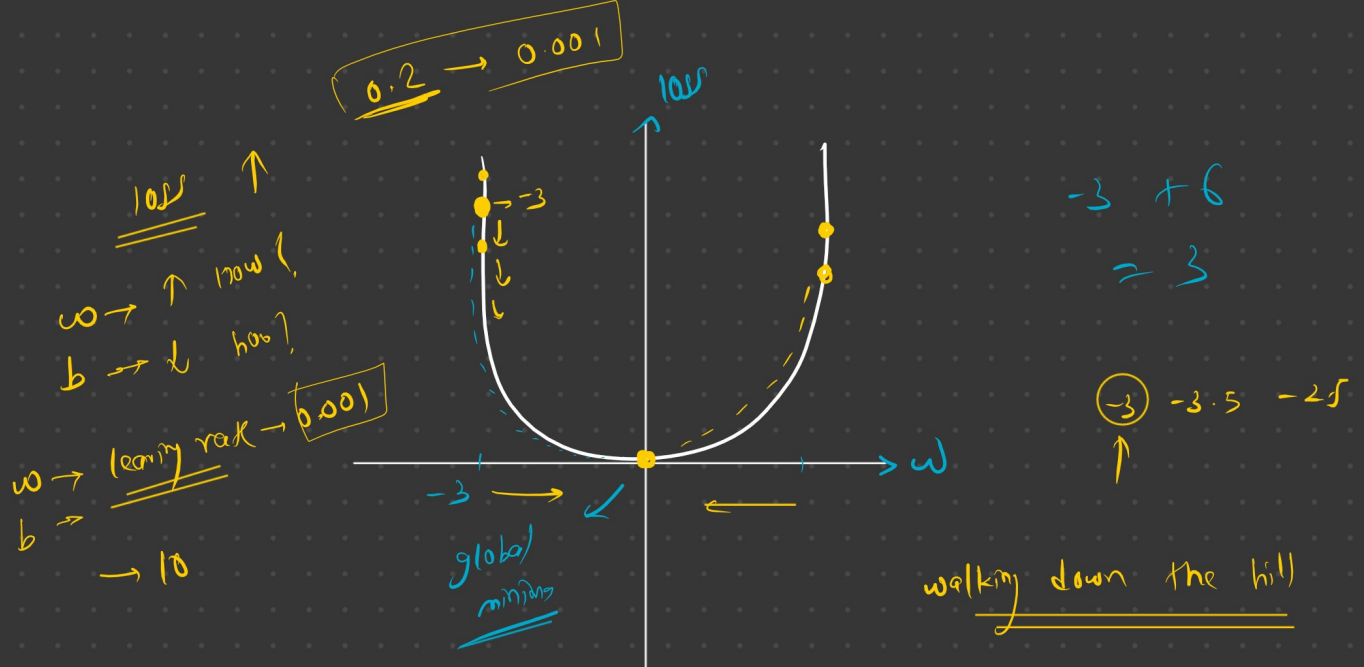
$$0.5 \cdot 2 + 1 = 2 \rightarrow \text{outcome}$$

$$\underline{1 + 1 = 2} \rightarrow 0$$



global minimum count $\rightarrow 1$

local minimum $\rightarrow \geq 0$



Top of hill
 • $\rightarrow$ constant - 1 - step $\rightarrow$ • ground $\rightarrow$ static learning rate
 • $\rightarrow$ running $\rightarrow$ jogging $\rightarrow$ walking $\rightarrow$ • ground Adaptive learning

Derivative

A derivative measure how a function change, as its input changes.
 It tells us "slope" or "rate of change" of a function at a given point.



Derivative tells us $\rightarrow$ "How steep is the curve of a function at a specific point & in which direction it is going"

let's say we have a function $\longrightarrow f(x) = x^2$
 $\rightarrow$ the graph of the function is parabola,
 $f(1) = 1^2 = 1$
 $f(2) = 2^2 = 4$

$\rightarrow$ Blue curve is the function

$$f(x) = x^2$$

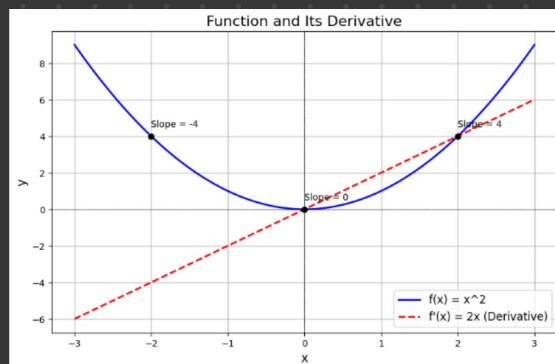
$\rightarrow$ Red line is the derivative

$$f'(x) = 2x$$

Formula

$$x^n \rightarrow nx^{n-1}$$

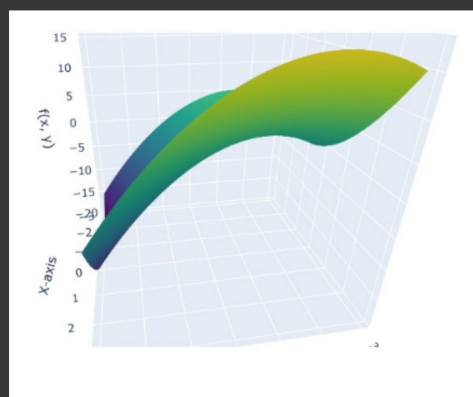
$$x^2 \rightarrow 2 \cdot x^1 \rightarrow 2x$$



$\rightarrow$ It crosses the x-axis at $x=0$, where the slope of $f(x) = 0$

$\rightarrow$ let's take an example of function with 2 variables,
in total 3 i.e. output of 2 variables will be $f(x, y)$

$$f(x, y) \rightarrow \underbrace{x^2 - y^2}_{\text{saddle}} + \underbrace{2x + 3y}_{\text{tilt the surface}}$$



$$\frac{\partial f}{\partial x} = 2x + 2 \rightarrow \text{rate of change in the } x \text{ direction}$$

$$\frac{\partial f}{\partial y} = -2y + 3 \rightarrow \text{rate of change in } y \text{ direction}$$

Gradient Descent

(1) Take the derivative of the loss function for each parameter in it.

$$\begin{aligned} &\rightarrow \text{o/p} \Rightarrow y = mx + c \rightarrow \text{+ve, -ve} \\ \text{loss} &\rightarrow \text{o/p} - \text{Actual} = \text{loss} \rightarrow \frac{\partial L}{\partial m}, \frac{\partial L}{\partial c} \end{aligned}$$

(2) Pick random values for the parameter

(3) plug parameter values into the derivative $\rightarrow$ o/p $\rightarrow$ slope

(4) calculate step size : $\text{step size} = \text{slope} \times \text{learning rate}$
 $\left(\frac{\partial L}{\partial p}\right) (1)$

(5) calculate new parameters :

$$m_{\text{new}} = m_{\text{old}} - \text{step size}$$

$$c_{\text{new}} = c_{\text{old}} - \text{step size}$$

(6) Repeat step 3, 4, 5 until you reach close to global minimum.

$\rightarrow$ Back propagation is responsible for computing gradients of the loss wrt each weight & bias in the network.

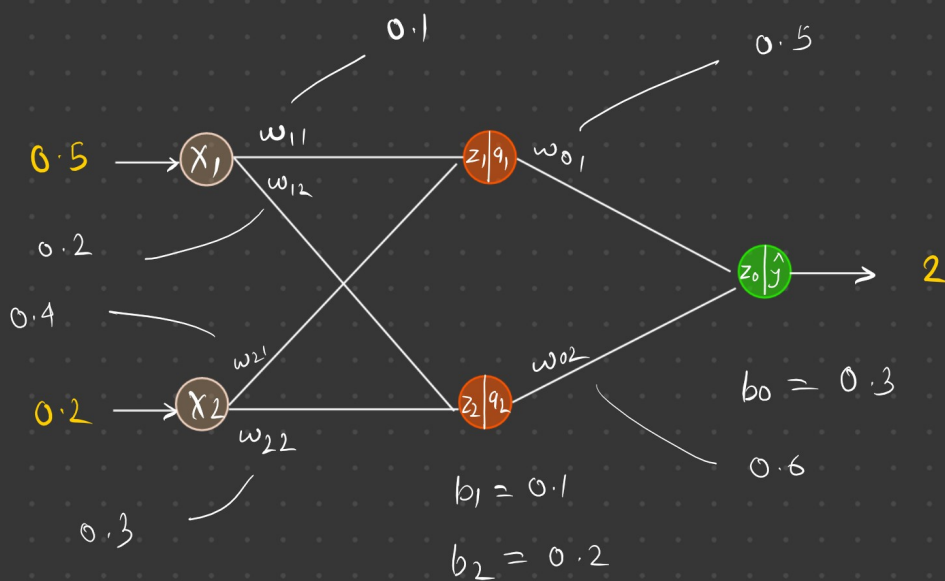
$\rightarrow$ once the gradients have been calculated via backpropagation, Gradient Descent uses these gradients to update the parameters which are weights & biases of network.

$$\text{new weight} = \text{old weights} - \eta \frac{\partial L}{\partial w} \xrightarrow{\text{BP}} \text{BP}$$

$\downarrow$
start

→ without Back propagation, we wouldn't know how to calculate the gradients for each w & b in the network.

→ without Gradient Descent, even if we know gradients, we wouldn't have an efficient way to update the parameters & minimize the loss.



Forward Pass 1

$$\boxed{z_1} = w_{11}x_1 + w_{21}x_2 + b_1 = (0.1)(0.5) + (0.4)(0.2) + 0.1 = 0.05 + 0.08 + 0.1 = 0.23$$

$$\boxed{z_2} = w_{12}x_1 + w_{22}x_2 + b_2 = (0.2)(0.5) + (0.3)(0.2) + 0.2 = 0.1 + 0.06 + 0.2 = 0.36$$

$$a_1 = \text{relu}(z_1) = \max(0, z_1) = \max(0, 0.23) = 0.23$$

$$a_2 = \text{relu}(z_2) = \max(0, z_2) = \max(0, 0.36) = 0.36$$

$$\begin{aligned}\hat{y} = z_0 &= w_{01} \cdot a_1 + w_{02} \cdot a_2 + b_0 = (0.5)(0.23) + (0.6)(0.36) + 0.3 \\ &= 0.115 + 0.216 + 0.3 \\ &= 0.631\end{aligned}$$

$$\hat{y} = z_0 = 0.631$$

Loss calculation

y - Actual output, $\hat{y} = z_0$ - predicted output

loss $\rightarrow$ MSE

$$\begin{aligned}\rightarrow \frac{1}{2}(\hat{y} - y)^2 &= \frac{1}{2}(0.631 - 2)^2 \\ &= \frac{1}{2}(-1.369)^2 = \frac{1}{2}(1.874)\end{aligned}$$

iteration 1 loss

$$= 0.937$$

Back propagation

$\rightarrow$ we will calculate gradient to update weights & biases.

o/p layer gradient:

$$\frac{\partial \text{loss}}{\partial \hat{y}} = \frac{1}{2}(\hat{y} - y)^2 = \frac{2}{2}(\hat{y} - y) = (\hat{y} - y) = -1.369$$

gradient of w_{01}, w_{02}, b_0 :

$q_1, w_{01}, q_2, w_{02}, b_0$



$$\frac{\partial \text{loss}}{\partial w_{01}} = \frac{\partial \text{loss}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_0} \cdot \frac{\partial z_0}{\partial w_{01}}$$

$$= (-1.369) \cdot (1) \cdot (0.23)$$

$$= \underline{\underline{-0.314}}$$

$$\frac{\partial \text{loss}}{\partial w_{02}} = \frac{\partial \text{loss}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_0} \cdot \frac{\partial z_0}{\partial w_{02}}$$

$$= (-1.369) \cdot (1) \cdot (0.36)$$

$$= \underline{\underline{-0.492}}$$

$$\frac{\partial \text{loss}}{\partial b_0} = \frac{\partial \text{loss}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_0} \cdot \frac{\partial z_0}{\partial b_0}$$

$$= (-1.369) \cdot (1) \cdot (1) = \underline{\underline{-1.369}}$$

$$\frac{\partial \text{loss}}{\partial q_1} = \frac{\partial \text{loss}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_0} \cdot \frac{\partial z_0}{\partial q_1} = (-1.369) \cdot (1) \cdot (0.5)$$

$$= \underline{\underline{-0.684}}$$

$$\frac{\partial \text{loss}}{\partial \hat{y}} = \underline{\underline{-1.369}}$$

$$\frac{\partial \hat{y}}{\partial z_0} = \hat{y} = z_0 = \frac{\partial \hat{y}}{\partial z_0} = \underline{\underline{1}}$$

$$\frac{\partial z_0}{\partial w_{01}} = z_0 = w_{01}q_1 + w_{02}q_2 + b_0$$

$$= \frac{\partial z_0}{\partial w_{01}} = q_1$$

$$= \underline{\underline{0.23}}$$

$$\frac{\partial z_0}{\partial w_{02}} =$$

$$z_0 = w_{01}q_1 + w_{02}q_2 + b_0$$

$$\frac{\partial z_0}{\partial w_{02}} = q_2 = \underline{\underline{0.36}}$$

$$\frac{\partial z_0}{\partial b_0} = w_{01}q_1 + w_{02}q_2 + b_0$$

$$\frac{\partial z_0}{\partial b_0} = \underline{\underline{1}}$$

$$\frac{\partial z_0}{\partial q_1} = w_{01}q_1 + w_{02}q_2 + b_0$$

$$= w_{01} = \underline{\underline{0.5}}$$

$$\frac{\partial z_0}{\partial q_2} = w_{01}q_1 + w_{02}q_2 + b_0$$

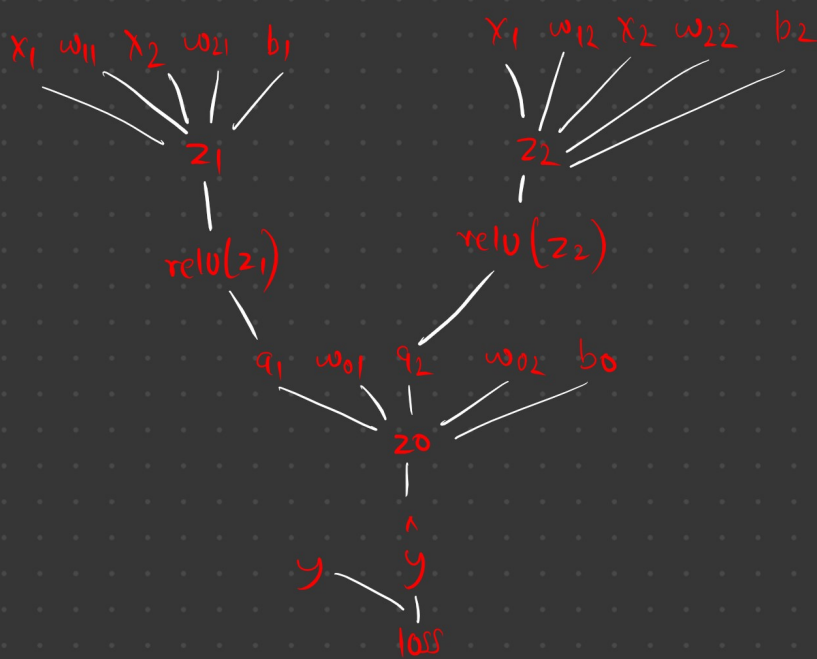
$$= w_{02}$$

$$= \underline{\underline{0.6}}$$

$$\frac{\partial \text{loss}}{\partial a_2} = \frac{\partial \text{loss}}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial z_0} \cdot \frac{\partial z_0}{\partial a_2}$$

$$= (-1.369) \cdot (1) \cdot (0.6)$$

$$= \underline{\underline{-0.821}}$$



calculating gradient of z_1 & z_2 :

$$\frac{\partial \text{loss}}{\partial z_1} = \frac{\partial \text{loss}}{\partial a_1} \cdot \frac{\partial a_1}{\partial z_1} = (-0.684) \cdot (1)$$

$$= \underline{\underline{-0.684}}$$

$$\frac{\partial \text{loss}}{\partial z_2} = \frac{\partial \text{loss}}{\partial a_2} \cdot \frac{\partial a_2}{\partial z_2} = (-0.821) \cdot (1)$$

$$= \underline{\underline{-0.821}}$$

$$\frac{\partial \text{loss}}{\partial a_1} = \underline{\underline{-0.684}}$$

$$\frac{\partial a_1}{\partial z_1} = a_1 = \max(0, z_1)$$

derivative of relu is

$$\frac{\partial a_1}{\partial z_1} = \begin{cases} x, & \text{if } z_1 > 0 \\ 0, & \text{if } z_1 \leq 0 \end{cases}$$

$$= \begin{cases} 1, & \text{if } z_1 > 0 \\ 0, & \text{if } z_1 \leq 0 \end{cases}$$

Since $z_1 = 0.23 > 0$

$$\frac{\partial a_1}{\partial z_1} = \underline{\underline{1}}$$

$$\frac{\partial \text{loss}}{\partial a_2} = -0.821$$

$$\frac{\partial a_2}{\partial z_2} = a_2 = \max(0, z_2)$$

derivative of relu is

$$= \begin{cases} x, & \text{if } z_2 > 0 \\ 0, & \text{if } z_2 \leq 0 \end{cases}$$

$$= \begin{cases} 1, & \text{if } z_2 > 0 \\ 0, & \text{if } z_2 \leq 0 \end{cases}$$

since $z_2 = 0.36 > 0$

$$\frac{\partial a_2}{\partial z_2} = 1$$

calculated gradient for $w_{11}, w_{21}, b_1, w_{12}, w_{22}, b_2$

$$\frac{\partial \text{loss}}{\partial w_{11}} = \frac{\partial \text{loss}}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_{11}} = (-0.684)(0.5)$$

$$= -0.342$$

$$\frac{\partial \text{loss}}{\partial w_{21}} = \frac{\partial \text{loss}}{\partial z_1} \cdot \frac{\partial z_1}{\partial w_{21}} = (-0.684)(0.2)$$

$$= -0.136$$

$$z_1 = w_{11}x_1 + w_{21}x_2 + b_1$$

$$\frac{\partial z_1}{\partial w_{11}} = x_1 = 0.5$$

$$\frac{\partial z_1}{\partial w_{21}} = x_2 = 0.2$$

$$\frac{\partial \text{loss}}{\partial b_1} = \frac{\partial \text{loss}}{\partial z_1} \cdot \frac{\partial z_1}{\partial b_1} = (-0.684)(1)$$

$$= \underline{\underline{-0.684}}$$

$$\frac{\partial \text{loss}}{\partial w_{12}} = \frac{\partial \text{loss}}{\partial z_2} \cdot \frac{\partial z_2}{\partial w_{12}} = (-0.821)(0.5)$$

$$= \underline{\underline{-0.4105}}$$

$$\frac{\partial \text{loss}}{\partial w_{22}} = \frac{\partial \text{loss}}{\partial z_2} \cdot \frac{\partial z_2}{\partial w_{22}} = (-0.821)(0.2)$$

$$= \underline{\underline{-0.1642}}$$

$$\frac{\partial \text{loss}}{\partial b_2} = \frac{\partial \text{loss}}{\partial z_2} \cdot \frac{\partial z_2}{\partial b_2} = (-0.821)(1)$$

$$= \underline{\underline{-0.821}}$$

$$\frac{\partial z_1}{\partial b_1} = \underline{\underline{1}}$$

$$z_2 = w_{12}x_1 + w_{22}x_2 + b_2$$

$$\frac{\partial z_2}{\partial w_{12}} = x_1 = \underline{\underline{0.5}}$$

$$\frac{\partial z_2}{\partial w_{22}} = x_2 = \underline{\underline{0.2}}$$

$$\frac{\partial z_2}{\partial b_2} = \underline{\underline{1}}$$

Gradient Descent

Now that the Back propagation is done & we have calculated gradients, we will do weight updation i.e. Gradient Descent

$$P_{\text{new}} = P_{\text{old}} - \eta \cdot \frac{\partial \text{loss}}{\partial \text{param}}$$

$\eta \rightarrow$ learning rate

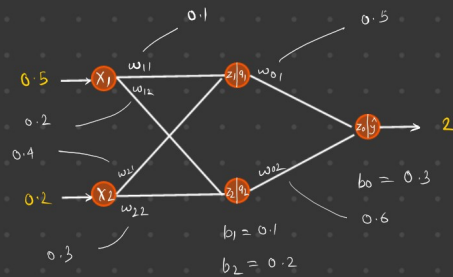
$$\eta = 0.02$$

$\partial \text{param} \rightarrow$ weights & biases.

$$\begin{aligned} w_{11_{\text{new}}} &= w_{11} - \eta \frac{\partial \text{loss}}{\partial w_{11}} \\ &= 0.1 - (0.02)(-0.342) \\ &= \underline{\underline{0.106}} \end{aligned}$$

$$\begin{aligned} w_{21_{\text{new}}} &= w_{21} - \eta \frac{\partial \text{loss}}{\partial w_{21}} \\ &= 0.4 - (0.02)(-0.136) \\ &= \underline{\underline{0.402}} \end{aligned}$$

$$\begin{aligned} w_{12_{\text{new}}} &= w_{12} - \eta \frac{\partial \text{loss}}{\partial w_{12}} \\ &= 0.2 - (0.02)(-0.410) \\ &= \underline{\underline{0.208}} \end{aligned}$$



$$\begin{aligned} w_{22_{\text{new}}} &= w_{22} - \eta \frac{\partial \text{loss}}{\partial w_{22}} \\ &= 0.3 - (0.02)(-0.184) \\ &= \underline{\underline{0.303}} \end{aligned}$$

$$\begin{aligned} b_{1_{\text{new}}} &= b_1 - \eta \frac{\partial \text{loss}}{\partial b_1} \\ &= 0.1 - (0.02)(-0.684) \\ &= \underline{\underline{0.113}} \end{aligned}$$

$$\begin{aligned} b_{2_{\text{new}}} &= b_2 - \eta \frac{\partial \text{loss}}{\partial b_2} \\ &= 0.2 - (0.02)(-0.821) \\ &= \underline{\underline{0.216}} \end{aligned}$$

$$w_{01_new} = w_{01} - \eta \frac{\partial loss}{\partial w_{01}}$$

$$= 0.5 - (0.02)(-0.314) = \underline{0.506}$$

$$w_{02_new} = w_{02} - \eta \frac{\partial loss}{\partial w_{02}}$$

$$= 0.6 - (0.02)(-0.492) = \underline{0.609}$$

$$b_{0_new} = b_0 - \eta \frac{\partial loss}{\partial b_0}$$

$$= 0.3 - (0.02)(-1.369) = \underline{0.327}$$

forward pass 2

$$\boxed{z_1} = w_{11}x_1 + w_{21}x_2 + b_1$$

$$= (0.106)(0.5) + (0.402)(0.2) + 0.113$$

$$= 0.246$$

$w_{11} = 0.106$
 $w_{21} = 0.402$
 $w_{12} = 0.208$
 $w_{22} = 0.303$
 $b_1 = 0.113$
 $b_2 = 0.216$
 $w_{01} = 0.506$
 $w_{02} = 0.609$
 $b_0 = 0.327$

$$\boxed{z_2} = w_{12}x_1 + w_{22}x_2 + b_2$$

$$= (0.208)(0.5) + (0.303)(0.2) + 0.216$$

$$= 0.381$$

$$\boxed{q_1} = \text{relu}(z_1) = 0.246$$

$$\boxed{q_2} = \text{relu}(z_2) = 0.381$$

$$\begin{aligned}\hat{y} = z_0 &= w_{01} \cdot a_1 + w_{02} \cdot a_2 + b_0 \\ &= (0.506)(0.246) + (0.609)(0.381) + 0.327 \\ &= 0.683\end{aligned}$$

$$\hat{y} = z_0 = 0.683$$

calculate loss $\rightarrow \frac{1}{2} (y_{\text{pred}} - y)^2$

$$\begin{aligned}\rightarrow \frac{1}{2} (0.683 - 2)^2 &= \frac{1}{2} (-1.317)^2 \\ &= \frac{1}{2} (1.734) \\ &= \underline{\underline{0.867}}\end{aligned}$$

old loss	new loss
0.937	0.867